

Conference Paper Title*

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Abstract—Although deep learning has considerably improved semantic communication, image transmission systems still face persistent challenges in accurately capturing the spatial position of the person in the image and background distortion of images in unmanned aerial vehicle UAV-based wilderness search and rescue scenarios. To address these issues, we propose a CAFB-YOLO-based semantic communication system with a semantic knowledge base for wilderness search-and-rescue scenarios. Specifically, we design an AF module to facilitate feature fusion and multi-scale feature interaction, resulting in an improved YOLO11 model, named CAFB-YOLO. CAFB-YOLO is incorporated into the semantic communication system to improve the accuracy of capturing spatial position of the person in image. To improve the quality of the reconstructed image avoiding background distortion of images, we design a background image knowledge base, and Stable Diffusion Inpainting is adopted to generate background images for the knowledge base. To enhance transmission reliability of important semantic information, diversity combining technique, which imaximal ratio combining (MRC), are incorporated into the system. Extensive experiments demonstrate that significantly improves image reconstruction quality, particularly under the low signal-to-noise ratio (SNR) conditions.

Index Terms—Semantic communication, YOLO, MRC, semantic knowledge base.

I. INTRODUCTION

Semantic communication has recently gained significant attention as a promising technology, which has attracted considerable attention from both academia and industry [1]. Unlike conventional communication that aims at the accurate delivery of symbols, the semantic communication targets the precise content reconstruction with equivalent semantics. It has the potential to achieve ultra-low compression rates and extremely high transmission efficiency. Semantic communication is considered the future of mobile communication, aiming to address bandwidth limitations in modern high-capacity multimedia content transmission.

In the field of image semantic communication, existing deep learning-based research can generally be classified into two directions. The first direction relies on unimodal features, where neural network architectures are optimized to obtain high-quality semantic representations of images [2], [3]. In addition, noise-robust modules have been designed to improve the accuracy of image reconstruction under noisy channel conditions. The second direction emphasizes multimodal feature fusion, where textual and visual information are jointly exploited to achieve richer semantic representations [4]. Compared with unimodal approaches, multimodal models generally achieve better reconstruction performance, particularly in unstable network environments. Despite achieving promising results in

certain scenarios, existing semantic communication models still face two critical challenges in complex human-centric scenarios: (1) the difficulty in accurately capturing the spatial position of the person in the image, which leads to distorted or unrealistic reconstructions; (2) semantic features are vulnerable to channel noise, which results in quality degradation during image transmission. In human-centric scenarios, such as wilderness search and rescue, accurate reconstruction of the coordinate and visual appearance of the target person is essential for reliable target identification and effective rescue coordination.

Therefore, to address the first challenge, we incorporate the bounding box coordinate of the target person as an auxiliary modality to complement visual features. Compared with textual descriptions, the bounding box coordinate provides an explicit representation of the spatial position in human-centric images. More generally, the bounding box coordinate can distinguish the boundary between the human target and the background. Therefore, the proposed framework is not limited to wilderness search and rescue scenarios and it can be naturally extended to other visual scenarios by replacing the bounding box region at the specified coordinates in the background image with the cropped image of target person. These coordinates and cropped images provide explicit spatial constraints to guide semantic encoding and preserve structural consistency during transmission.

To address the second challenge, we introduce a background image knowledge base [5] and conventional wireless communication techniques to mitigate the impact of noise on semantic features. The background image knowledge base serves as a repository of the background images. By referencing a specific background image from the knowledge base, communication systems can more robustly infer and reconstruct the intended information, even when the transmitted data is corrupted or incomplete. Stable Diffusion Inpainting is adopted to generate the background images for this knowledge base. The background image knowledge base provides background information to recover the visual content of non-critical regions outside the target person, thereby making the reconstructed image more realistic. Furthermore, considering that different semantic features have different importance, conventional wireless communication techniques are incorporated to provide differentiated transmission reliability. Specifically, Adaptive Coding and Modulation (ACM) [6] and diversity combining techniques [7] are employed to improve transmission reliability and spectral efficiency.

Therefore, in this paper, we propose a knowledge-base-

assisted semantic communication system for wilderness search and rescue scenarios. Specifically, we employ YOLOv8 to extract semantic features and reduce redundancy. By incorporating conventional wireless communication techniques and using the background image as additional auxiliary knowledge, we achieve superior image reconstruction performance, especially under challenging channel conditions. The major contributions of this paper are summarized as follows:

- We design a knowledge-base-assisted semantic communication system that enables the network to capture the spatial position of the person in the image often overlooked by prior approaches, thereby achieving higher-quality image reconstruction.
- We propose a novel background image knowledge base, which differs from prior knowledge base designs based on textual annotations or category-level semantic vectors. With background information through background images, the background image knowledge base provides background priors that directly complement visual semantics and significantly enhance decoding under challenging channel conditions.

II. SYSTEM MODEL

The proposed semantic communication system for wilderness search and rescue scenarios, as depicted in Fig. 1, consists of semantic encoder, transmission scheme, semantic decoder, and background image knowledge base. The background image knowledge base will be introduced in section III.

A. Rician Channel

The Rician fading channel is a radio propagation model used when the received signal has both a strong direct path and reflected multipath. The equation of the Rician are described as follows

$$\mathbf{h} = \sqrt{\frac{K}{K+1}} \mathbf{h}_{LOS} + \sqrt{\frac{1}{K+1}} \mathbf{h}_{NLOS}, \quad (1)$$

where $\mathbf{h}_{LOS}$ is the LOS component and $\mathbf{h}_{NLOS}$ is the NLOS component that is given by the Rayleigh fading channel model, which can be defined as:

$$\mathbf{h}_{NLOS} = \mathbf{h}_1 + j\mathbf{h}_2, \quad (2)$$

where $\mathbf{h}_1 \sim \mathcal{N}(0, \frac{\sigma^2}{2})$ and $\mathbf{h}_2 \sim \mathcal{N}(0, \frac{\sigma^2}{2})$, therefore, $\rho \sim \mathcal{CN}(0, \sigma^2)$.

B. Semantic Encoder

The source image is encoded into two distinct semantic features [8], namely the bounding box coordinate of the target person and the cropped image of the target person from the source image, utilizing two semantic extractors [9] based on the YOLOv8 model and down sampling [10]. The bounding box coordinate is extracted only using YOLOv8 model, the cropped image is extracted using YOLOv8 model with down sampling for further compression. For simplicity, we use subscripts 1 and 2 to replace subscripts coordinate and cropped

image of person, respectively, in the following discussion. The i th extracted feature can be expressed by

$$\mathbf{S}_i = \mathcal{E}_i(\mathbf{X} | \boldsymbol{\theta}_i^*), \forall i \in \{1, 2\} \quad (3)$$

where $\mathcal{E}_i(\mathbf{X} | \boldsymbol{\theta}_i^*)$ is the i th model with $\boldsymbol{\theta}_i^*$ being network parameters and $\mathbf{X}$ is the source image.

To align with existing digital communication systems, the semantic feature $\mathbf{S}_i$ is transformed into a bit sequence $\mathbf{K}_i$, which can be expressed as

$$\mathbf{K}_i = \mathcal{B}(\mathbf{S}_i), \quad (4)$$

where $\mathcal{B}(\cdot)$ is a binary mapping function.

C. Transmission Module

Due to the different importance of the semantic data streams, multi-stream transmissions are considered in our system. The received data streams are expressed as

$$[\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2] = \mathcal{T}([\mathbf{K}_1, \mathbf{K}_2]), \quad (5)$$

where $\mathcal{T}(\cdot)$ is the transmission scheme, which comprises diversity/combining, the channel coding/decoding and modulation/demodulation components.

In the transmission scheme, the bit sequences $\mathbf{K}_1$ and $\mathbf{K}_2$ are both encoded using LDPC channel coding, but with different code rates and modulation schemes [11]. The bit sequence $\mathbf{K}_1$ is transmitted using low-order modulation, low rate channel coding, and diversity combining scheme, whereas $\mathbf{K}_2$ is conveyed via an adaptive coded modulation scheme.

Rayleigh fading channel with additive white Gaussian noise [12] is considered as the channel model in this paper to realistically simulate the wireless transmission environment in wilderness search and rescue scenarios.

D. Semantic Decoder

In the semantic decoder [13], the received semantic data streams $\hat{\mathbf{K}}_i$ are first reconverted to the semantic features $\hat{\mathbf{S}}_i$, which can be expressed by

$$\hat{\mathbf{S}}_i = \mathcal{B}^{-1}(\hat{\mathbf{K}}_i), \quad (6)$$

where $\mathcal{B}^{-1}(\cdot)$ is the inverse operation of $\mathcal{B}(\cdot)$. After obtaining the semantic features $\hat{\mathbf{S}}_i$, a background image $\mathbf{U}$ is required for further processing. The background image $\mathbf{U}$ is provided from the knowledge base generated by Stable Diffusion Inpainting model. Finally, the semantic features $\hat{\mathbf{S}}_i$ and the background image $\mathbf{U}$ are forwarded to the image reconstruction D to reconstruct $\hat{\mathbf{X}}$, which can be expressed as

$$\hat{\mathbf{X}} = D(\hat{\mathbf{S}}_1, \hat{\mathbf{S}}_2, \mathbf{U}) = \mathcal{D}(\hat{\mathbf{K}}_1, \hat{\mathbf{K}}_2, \mathbf{U}), \quad (7)$$

where $\mathcal{D}$ is the semantic decoder.

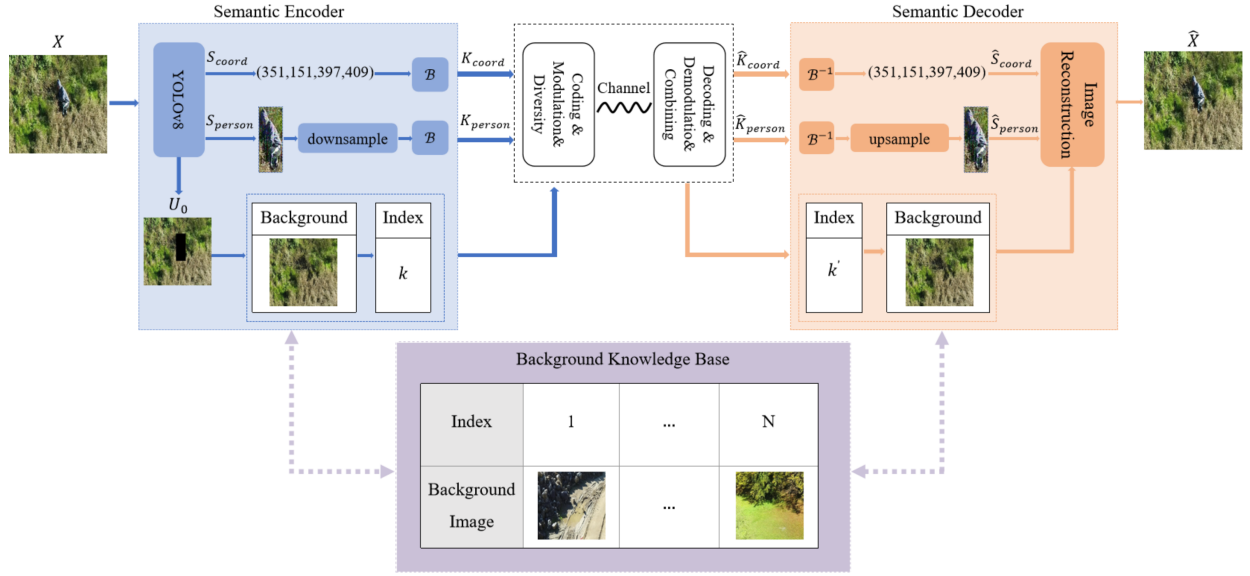


Fig. 1. The proposed semantic communication system for wilderness search and rescue scenarios.

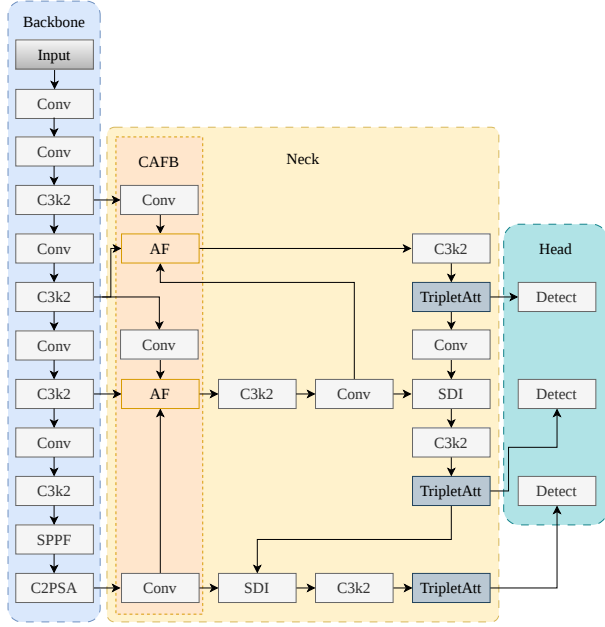


Fig. 2. The CAFB-YOLO model structure.

III. ALGORITHM

A. cAFB-YOLO

As shown in Fig. 2, the CAFB-YOLO model structure can be divided into three parts: Backbone, Neck and Head. Different from YOLO11, we improve the Neck by designing the AF module and the CAFB architecture.

In addition, the Triplet Attention mechanism introduced in Reference [7] and the SDI module are incorporated into the proposed model structure.

Reference [7] introduces the Triplet Attention mechanism in the neck feature fusion layer. Different from the module that

only focuses on channel information or the Spatial Attention that only focuses on spatial information, the Triplet Attention establishes cross-dimension interaction dependencies through three branches without dimensionality reduction operations, thus avoiding the loss of feature information.

However, the performance improvement achieved by directly applying Triplet Attention to the YOLO11 model is not significant. Therefore, based on this mechanism, we further design the CAFB architecture. Specifically, four feature layers from the Backbone are introduced into the Neck, and the proposed AF module is employed to achieve feature fusion and the interaction of multi-scale feature information.

The proposed AF module is shown in Fig. 3, AF integrates channel information and spatial information from three feature scales, three input feature are represented as $\mathbf{X}_l \in \mathbb{R}^{C_l \times H_l \times W_l}$, $\mathbf{X}_m \in \mathbb{R}^{C_m \times H_m \times W_m}$, $\mathbf{X}_s \in \mathbb{R}^{C_s \times H_s \times W_s}$, where $\mathbf{X}_l$ denotes the large-scale feature, $\mathbf{X}_m$ denotes the middle-scale feature, $\mathbf{X}_s$ denotes the small-scale feature. After processing through the AF module, the feature representation can be expressed as

$$\mathbf{F} = \text{Concat}(\mathcal{P}_{max}(\mathbf{X}_l) + \mathcal{P}_{avg}(\mathbf{X}_l), \mathbf{X}_m, \mathcal{U}(\mathbf{X}_s)) \quad (8)$$

where $\mathcal{P}_{max}(\cdot)$ denotes the adaptive maximum pooling operation, $\mathcal{P}_{avg}(\cdot)$ denotes the adaptive average pooling operation, $\text{Concat}(\cdot)$ denotes fusion in the channel direction, and $\mathbf{F} \in \mathbb{R}^{C_m \times H_m \times W_m}$.

By integrating high-level semantic information, middle-level information, and low-level spatial details, the proposed module effectively enhances multi-scale feature representation capability, thereby improving detection performance, especially for small objects and wilderness scenes.

B. Background Image Knowledge Base

The proposed background image knowledge base is applied to the reconstruction of target person images in wilderness

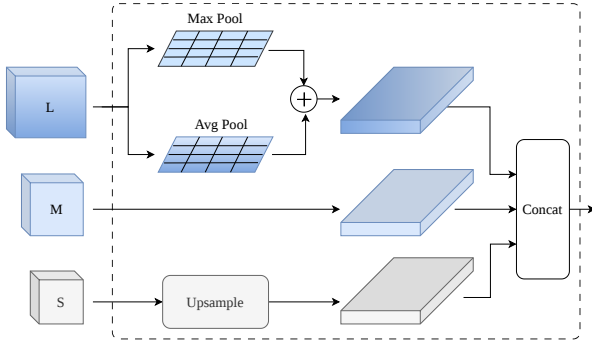


Fig. 3. The AF module.

search and rescue scenarios. The background image knowledge base provides a background image as input to the image reconstruction of semantic decoder.

Stable Diffusion Inpainting is a latent text-to-image diffusion model [14] that can generate photo-realistic images from text input and can also perform inpainting by using a mask.

In this paper, the Stable Diffusion Inpainting is used to generate a background image U in the background image knowledge base. To improve its performance, several improvements are applied to the model inputs. The Stable Diffusion Inpainting model requires three inputs, including a text prompt, an input image, and a corresponding mask image. The text prompt can be manually set to an appropriate value according to the wilderness search and rescue scenarios, and it has little impact on the output U . The mask image is obtained by directly processing the input image with a pretrained background removal model, namely rembg.

IV. EXPERIMENTS

We conducted experiments on the SARD dataset, with a resolution of 640×640 . The experiments include the training and testing of AF-YOLO model, as well as simulations of the semantic communication system.

A. AF-YOLO

The original SARD dataset contains 5755 images of wilderness search and rescue scenarios. 4041 images were selected as the training set, 1144 images as the validation set and 570 images as the test set.

We train detectors with GPU (NVIDIA RTX 3050), adopting the stochastic gradient descent (SGD) optimizer with the initial learning rate 0.001, and cosine annealing method is used to reduce the learning rate. The batch size is 16, the total number of training rounds (epochs) is 100

In order to evaluate the detection effectiveness of AF-YOLO model algorithm, the following evaluation metrics are used in this paper: mean Average Precision(mAP 50-95 and mAP 50), and Inference Time. A comparison was conducted between AF-YOLO and the smaller series of models within the YOLO11 series. The results are summarized in Table I, it is evident that AF-YOLO surpasses YOLO11 in terms of accuracy

TABLE I
PERFORMANCE VERSUS THE MODEL.

Model	mAP@0.5	mAP	Inference Time (ms)
YOLO11	0.918	0.553	8.9
TripleAtt-YOLO	0.908	0.554	11.0
AF-YOLO	0.923	0.581	12.5

B. Semantic Communication System Performance

Semantic communication system simulations are conducted on the test set consisting of 570 images. We evaluated the model on Rician channel. The SNR was varied from 0 dB to 16 dB to simulate different channel conditions. The modulation and coding schemes of these two semantic data streams are different. The semantic data stream K_1 employs the diversity and combining techniques MRC with 3 receiving branches with $L = 3$, whereas K_2 does not. The coordinate data streams employ LDPC coding with a rate of 1/4 and BPSK modulation, but the cropped person image data streams employ LDPC coding with a rate of 1/4 and BPSK modulation. Monte Carlo simulation approach is used in simulation of transmission stage.

To evaluate the performance of the proposed model, we compare it with several baseline methods, including [12], Pose-VAE without SKB, DeepJSCC [16], and the traditional BPG+LDPC pipeline. We adopt three widely used metrics to quantitatively assess the quality of the generated images: LPIPS, SSIM, and PSNR.

Fig. 3 compares the performance of the proposed method with baseline schemes over an AWGN channel in terms of PSNR, SSIM, and LPIPS. As shown, consistently outperforms all baseline models across the entire SNR range. This performance gain is mainly attributed to the incorporation of pose information, which enables more accurate modeling of image spatial structure. Moreover, the improvement is more pronounced under low-SNR conditions, where the Pose-SKB provides prior structural.

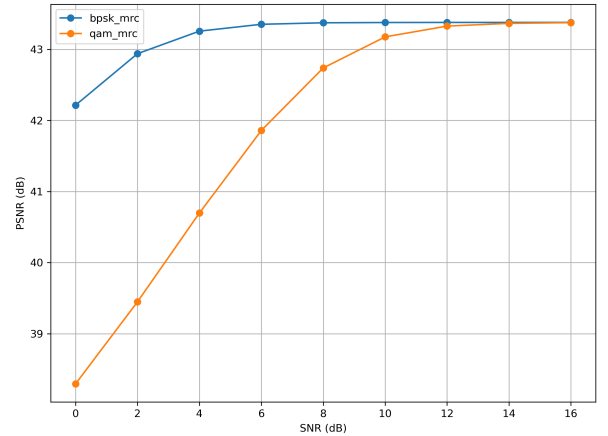


Fig. 4. The BER performance of the cropped person image under different coded modulation schemes.

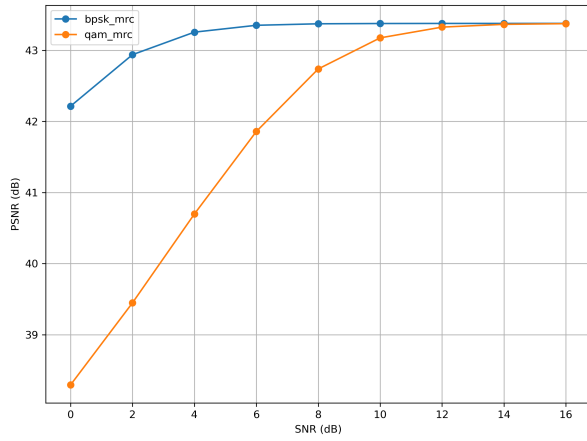


Fig. 5. The BER performance of the total under different coded modulation schemes.

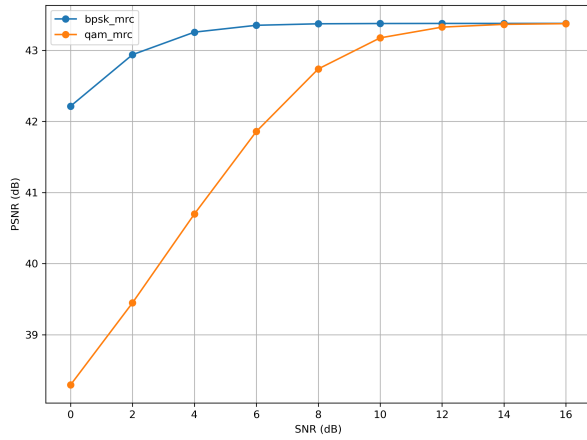


Fig. 6. The BER performance of the total under different combining strategies.

Figs. ??–?? presents a qualitative comparison of the image reconstruction results under various SNRs for the coded modulation scheme CMS_N. It can be observed that all six coded modulation schemes achieve relatively high image quality. Even at low SNR, the target can still be accurately localized, which is of great importance in wilderness search and rescue scenarios.

V. CONCLUSION

This letter proposes a semantic communication system model named Pose-VAE for image transmission. By incorporating human pose information, Pose-VAE is able to better, the system transmits only the critical semantic information, while the remaining non-essential information is recovered at the receiver based on prior knowledge stored in a knowledge base, simplifying the communication process and reducing storage requirements. By introducing ACM, diversity and combining techniques, our system effectively improves BER performance by transmitting semantic information at higher data rate and certain transmission quality, while decreasing the coordinate

bit error rate of images, thereby improving image quality. Simulation results demonstrate that the semantic communication system employing the proposed method achieves performance improvements. For wilderness search and rescue scenarios, the target can still be accurately localized even in challenging channel with significant noise interference.

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